UNIVABIO · AI FOR HUMAN HEALTH

SafeStep

A silent guardian that reduces repetitive caregiving checks while helping older adults retain independence.

THE HEALTH PROBLEM

Caregivers need reliable signals about meals, medication, movement, and falls. Repeated calls and surveillance create anxiety for caregivers and make older adults feel micromanaged.

THE SOLUTION

SafeStep turns routine care activity into clear, reviewable signals. It combines a simplified senior interface with a caregiver dashboard, automated reminders, location support, activity verification, and fall alerts.

Respectful monitoring

Computer vision verifies selected activities without requiring a caregiver to watch a live feed.

Faster escalation

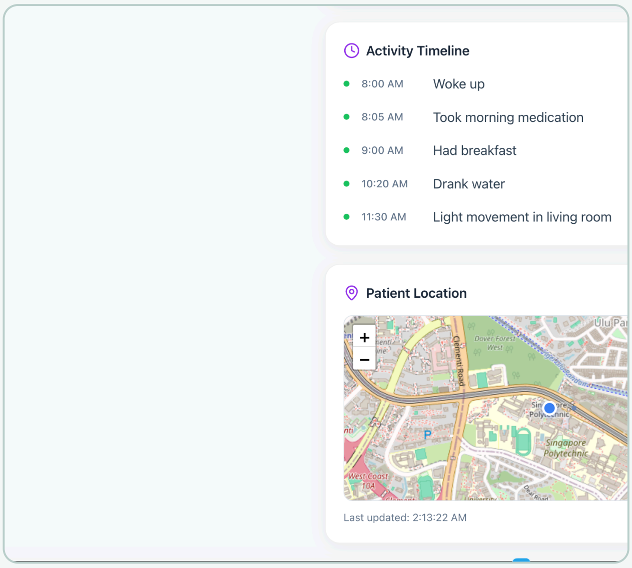
Pose and motion signals identify possible falls and surface high-confidence events.

Daily reassurance

Meal and medication events reduce the need for repeated “did you?” calls.

Useful context

Care summaries make routines and exceptions easier to review.



Prototype caregiver view showing activity and location context.

WORKING EVIDENCE

Video: youtube.com/watch?v=KopJA4DhsWY
Code: github.com/Ducksss/hack4good
Project: devpost.com/software/safesteps-9x4o7i

WHY IT FITS UNIVABIO

- Addresses a real human-health problem affecting older adults and caregivers.
- Combines AI, computer vision, reminders, maps, and caregiver software.
- Designs separate interfaces for seniors and caregivers.
- Separates demonstrated functionality from claims needing clinical validation.

TECHNICAL FOUNDATION

Interface: React 18, TypeScript, Tailwind CSS
Services: Node.js, Express, Firebase, MongoDB Atlas
AI / vision: Gemini, YOLOv8, MediaPipe, OpenCV, Tesseract
Care loop: Google Maps, Twilio

TEAM

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Pre-existing work disclosure: SafeStep is authorized team work created before UnivaBio and previously entered in other hackathons. This entry reuses its existing concept, code, interface, repository, video, screenshots, and documentation. It does not claim that the project was built during this event. SafeStep is a prototype and is not presented as a clinically validated medical device.